

A typical example of tolerant clustering with a relational constraint is the regionalization problem, in which we combine basic administrative geographical units into a smaller number of clusters or regions composed of similar and geographically connected units.

An example of leader clustering is identifying thematic clusters of authors in a weighted citation network between authors. The figure shows a cluster of authors from the field of traditional clustering, with Ward as the leader, based on the bibliography of clustering and block modeling. This cluster includes "classics" such as Rand, Johnson, Rohlf, Hartigan, Gordon, Hubert, and DeSarbo, among others.

Clustering with a relational constraint was introduced by Ferligoj and Batagelj [7, 8]. They also proposed solution procedures based on a dissimilarity matrix. Both agglomerative and local optimization approaches can be adapted. However, these procedures are not suitable for networks with a very large number of units (over 10,000). Large networks are typically sparse, and by limiting ourselves to dissimilarities only between linked units and to criterion functions such as minimum, maximum, or average, we can develop much more efficient procedures [5].

Procedures for clustering with relational constraints are partially supported in Pajek, a program for analysis and visualization of large networks [4]. The Pajek or netsJSON format [1] is used to describe network data. cluRC is an R package that offers both types of procedures to users. In this talk, we will present these procedures and demonstrate their application to real-world data. The cluRC package is available on GitHub [9].

References

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